

A
Synopsis
On

"ASSOCIATE QUALITY ASSURANCE"

Submitted to: Panjab University, Chandigarh in partial fulfillment of the requirement for the degree of Master of Computer Applications (MCA)

Submitted by:

Plaksha Maini

Roll-No: 28



Under the Guidance of

Dr. Rohini Sharma

**DEPARTMENT OF COMPUTER SCIENCE AND
APPLICATIONS**

PANJAB UNIVERSITY, CHANDIGARH

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Certificate from the Supervisor

This is to certify that Plaksha Maini, Roll Number 28, MCA 2nd Year (Morning), is undergoing training in "Software Testing(Associate Quality Assurance)" at "Chicmic Studios", Sector 74 phase 8-B, Mohali, under my supervision. The learning process involves understanding Manual And Automation Testing. This training is part of her academic curriculum.

Supervisor Name: DEVINDER KUMAR

Qualification: BTECH

Designation: TEAM LEADER

Employing Organization: "Chicmic Studios", Sector 74 phase 8-B, Mohali

Contact Information: devinder.kumar@chicmicstudios.in

Devinder Kumar
3/10/2025
(Signature with Date)

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Scope of Work

- Ensure that the software application meets all test-based performance and security requirements.
- Identify and resolve defects early in the development lifecycle.
- Automate repetitive and regression testing tasks to improve testing efficiency and reduce time-to-market.
- Provide comprehensive test coverage to ensure that all aspects of the software are tested under real-world conditions.
- Create and document test cases to verify functional requirements and edge cases.
- Perform exploratory testing to identify potential issues that may not be captured by predefined test cases.
- Log and track defects in a defect tracking system (e.g., JIRA, Mantis).
- Identify the test cases suitable for automation based on the project's needs (e.g., regression tests, repetitive test cases).

Project Outline

Background: Quality Assurance (QA) and Software Testing are critical elements in the software development lifecycle. Their primary goal is to ensure the delivery of high-quality, reliable, and secure software products to end-users. QA involves systematic activities designed to ensure the development process meets predefined quality standards, while software testing identifies and rectifies defects, ensuring functionality, performance, and security.

This training focuses on Expertise in Manual And Automation testing.

Manual Testing involves human testers executing test cases, verifying software functionality, identifying defects, and ensuring the system meets business requirements. This approach is crucial for exploratory testing, usability assessments, and areas where human intuition and judgment are required.

Automation Testing uses specialized tools and scripts to automatically execute predefined test cases, speeding up the testing process, especially for repetitive tasks like regression testing, performance testing, and large-scale testing. Automation will learn Selenium and how to write test scripts.

Scope of Work:

- Ensure that the software application meets all functional, performance, and security requirements.
- Identify and resolve defects early in the development lifecycle.
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- Provide comprehensive test coverage to ensure that all aspects of the software are tested under real-world conditions.
- Create and document test cases to verify functional requirements and edge cases.
- Perform exploratory testing to identify potential issues that may not be captured by predefined test cases.
- Log and track defects in a defect tracking system (e.g., JIRA, Mantis).
- Identify the test cases suitable for automation based on the project's needs (e.g., regression tests, repetitive test cases).

- Select appropriate test automation tools (e.g., Selenium, JUnit, TestNG, Appium, etc.) based on the project type and technology stack.
- Develop reusable and maintainable automation test scripts.

Objectives:

- Develop and execute both manual and automated test cases to cover all functional, non-functional, and edge case scenarios to ensure the software meets all requirements.
- Identify, log, and resolve defects during both manual and automated testing processes, ensuring that the software is free from critical bugs and operates as intended across various environments
- To report the issues on Bug Tracking Application(e.g Mantis).
- Use manual testing for exploratory and complex scenarios that require human judgment, while utilizing automation for repetitive and large-scale testing tasks.
- Verify that the software functions (responsive testing) seamlessly across different platforms (Windows, macOS, mobile OS) and browsers (Chrome, Firefox, Safari, etc.) through both manual and automated testing.
- Implement both manual and automated testing early in the development lifecycle to catch defects before they reach production, reducing the risk of critical issues in live environments.
- Ensure that automated test scripts are maintainable and easy to update in the face of changing application requirements, minimizing future efforts for script revisions.
- Use manual testing for user experience (UX) validation by performing pixel-to-pixel testing to ensure the application's usability aligns with user expectations and business requirements.

Plan of Work:

Phase 1: ManualAutomation

Objective:To learn about Manual testing General concepts

- What is software testing.
- Understanding Product VS Service Based Company
- Black box Testing

- White Box Testing
- Fundamental Test Process
- SDLC Model Basics
- What is STLC
- SDLC VS STLC
- Entry and exit criteria

Phase 2: Types of testing and what are test scenarios

Objective: Different types of functional and non functional testing

- **Types of testing(functional and non Functional)**
 - Different testing types
 - Static and dynamic testing
 - Smoke vs sanity testing
 - Monkey, Gorilla vs Glass Box testing
 - Regression testing
 - Cross Browser testing
 - Difference between regression and retesting
 - Compatibility and Responsive testing
 - What are test design techniques(Equivalence partitioning , Boundary Value analysis)
- **Test case plan**
- **Test Scenario with real time projects**
- **Test cases and test case templates**
- **Use cases**

Phase 3: Bug Tracking and Reporting

Objective: How to Differentiate bug impact on the projects and types of bugs.
Reporting the bugs on bug tracking platforms

- **Priority and it's Classifications**
- **Priority vs Severity**
- **Bug Life cycle**
- **Live project Testing**
- **Understanding Bug/Defect/Issue/Error/Mistake/Fail/**
- **Invalid bug-Why and how to avoid?**
- **Not reproducible bug- Why and how to avoid?**

- Bug reporting
- Build extension, Build Version
- Bug tracking tools(mantis,bugzilla,Jira)
- Bug reporting on Mantis.

Phase 4: Test levels in Software testing and types of software and applications

Objective: Learn levels of software testing.

- What is test levels?
- What is Unit/Component level testing?
- Unit/Component level testing practical example
- Integration testing types
- User Acceptance testing
- Automation enhancement
- Basics of a software/Applications
- Standalone application
- Single tier ,two and three tier applications architecture
- Understanding client, server and Test environment
- Live project testing

Platform to be Used:

- Different browsers,MAC ,Windows and mobile devices(Android ,IOS) for testing
- Mantis,Jira,Bugzilla For reporting issues

References:

1. testing types - Guru 99
2. Javatutorialspoint
3. Software testing help (website)

Tasks Performed:

- **General Scenario based test cases:** Let's practice writing test cases for a **morning routine** scenario. Here's the scenario:

Scenario: A person wakes up at 7:00 AM, checks their phone, makes coffee, brushes teeth, gets dressed, has breakfast, and leaves by 8:30 AM.

Your Task:

Write test cases for this morning routine, covering:

1. **Normal flow** (everything goes as planned)
2. **Edge cases** (e.g., alarm fails, coffee machine breaks, skips brushing teeth)
3. **Alternate paths** (e.g., skips breakfast, works from home)
4. **Test Case Structure:**
5. **Preconditions**
6. **Test Steps**
7. **Expected Results**
8. **Edge Cases**

- **Techrow Website and Application:** This was an international client project aimed at creating a platform with distinct roles for admins, students, teachers, and organizations. The platform enables the creation of quizzes and provides AI-driven services for organizations to offer to students. It also facilitates content creation by teachers for student engagement.

Key Learnings:

- Gained experience in **exploratory testing**, enhancing the ability to identify issues and ensure the flow of the project is smooth.
- Conducted **iOS and Android testing** to ensure cross-platform compatibility and optimal performance.
- Focused on **functional and UI testing** to validate the functionality and usability of the application.

From: Plaksha Maini

- **Mejhool Application:**Mejhool is a decentralized application currently working in its Beta version. The company is developing both hybrid and native applications. The web application is built using React, while the mobile applications are developed with React Native. The app also incorporates **notification handling** for user engagement and updates.

Key Learnings:

-Wrote **comprehensive test cases** for the entire application, covering various use cases and features.

-Conducted **testing and reported bugs** to ensure the application meets quality standards and performs as expected.

-Developed test cases for both **1-on-1 user chats** and **group chats**, ensuring proper functionality of chat features.

- **Speltipset Website:**Speltipset is a casino-focused website featuring different portals for Admin, Publishers, Operators, and Stores. Operators create various stores, while Publishers are assigned to manage each store, all under the supervision of the Admin. The platform enables seamless management of these roles and their respective responsibilities.

Key Learnings:

-Conducted **responsive testing** across different browsers and devices to ensure compatibility and optimal user experience.

-Performed **pixel-to-pixel testing** to verify design accuracy and alignment.

- Utilized various **responsive testing tools** to ensure the website adapts seamlessly to multiple screen sizes and resolutions.